

Vital Sleep Patch:

Transforming Pediatric Sleep Apnea Diagnosis through Home-Based Testing

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Interviews Conducted: 10

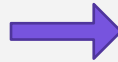
Problem Solving Journey: Started with a wearable sensor patch and AI-based diagnostic pipeline for pediatric OSA. Now, through customer discovery, we are focusing on understanding who actually values this, why they value it, and what evidence we need to validate that value proposition.



The Vital Sleep Patch offers a paradigm shift:

Problem:

In-clinic polysomnography is failing to address pediatric sleep apnea nationwide



Solution:

A send-home wearable sensor patch and AI-based diagnostic pipeline.



Stakeholders (and core problem faced)

End Users

Parents of children

Child sleep testing is slow, disruptive, and stressful for both the child and family.

Adults

Sleep issues are underdiagnosed because testing is inconvenient, slow, and hard to access.

Influencers / Referrers

Obesity Clinic

Sleep-related issues can go unidentified, delaying weight-loss treatment and care planning.

Orthodontist

Limited visibility into sleep-disordered breathing makes airway-focused treatment decisions harder.

Other Private Buyers (e.g.: Longevity)

Early sleep problems are hard to detect with clinical-grade confidence before they affect long-term health.

Offline Test Providers

AMC Sleep Medicine Clinic

Pediatric sleep testing is capacity-constrained, labor-intensive, and bottlenecked by in-lab studies.

Independent Test Centers

Throughput is limited by overnight lab capacity, staffing, and cumbersome testing workflows.

Service Providers

Tech-Enabled Service Providers & IDTFs

Current pediatric sleep testing options are hard to scale due to poor form factor, weak coverage, and fragmented workflows.

Device Manufacturers / OEMs

They lack strong pediatric-ready partners and products to extend into sleep diagnostics and care delivery.

Payors

Insurer

Expensive lab-based testing and delayed diagnosis increase downstream cost and inefficiency.

Vital Health

Stakeholders (and potential value prop)

End Users

Parents of children

Fast, at-home diagnosis to restore child's health and family sleep.

Adults

Fast, at-home diagnosis to restore their own health and sleep quality.

Influencers / Referrers

Obesity Clinic

Better referral triage to unblock weight-loss treatment pathways

Orthodontist

Screening to inform airway-focused orthodontic treatments.

Other Private Buyers (e.g.: Longevity)

Early detection of sleep issues for long-term developmental health.

Offline Test Providers

AMC Sleep Medicine Clinic

Efficient triage and increased throughput to manage backlogs.

Independent Test Centers

Expands capacity and throughput for sleep clinic overflow.

Service Providers

Tech-Enabled Service Providers & IDTFs

Unlocks pediatric market and improves their platform value to clinicians and families.

Device Manufacturers / OEMs

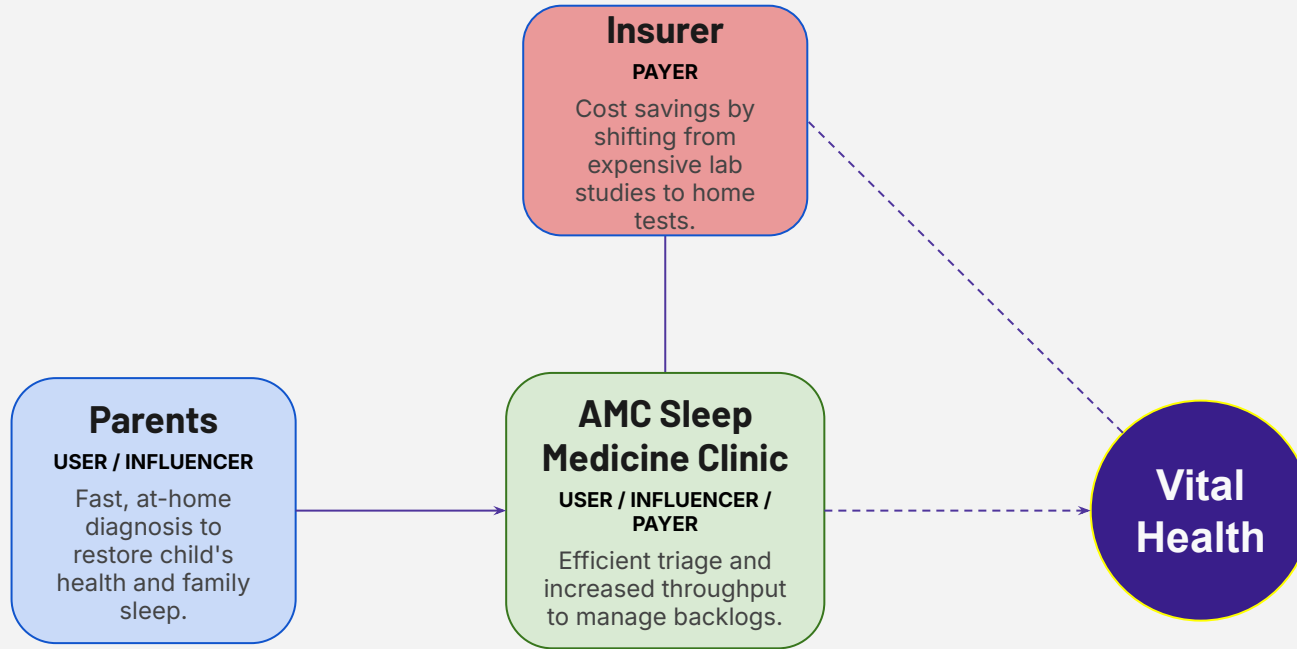
Unlocks potential partnerships for enhanced care delivery

Payors

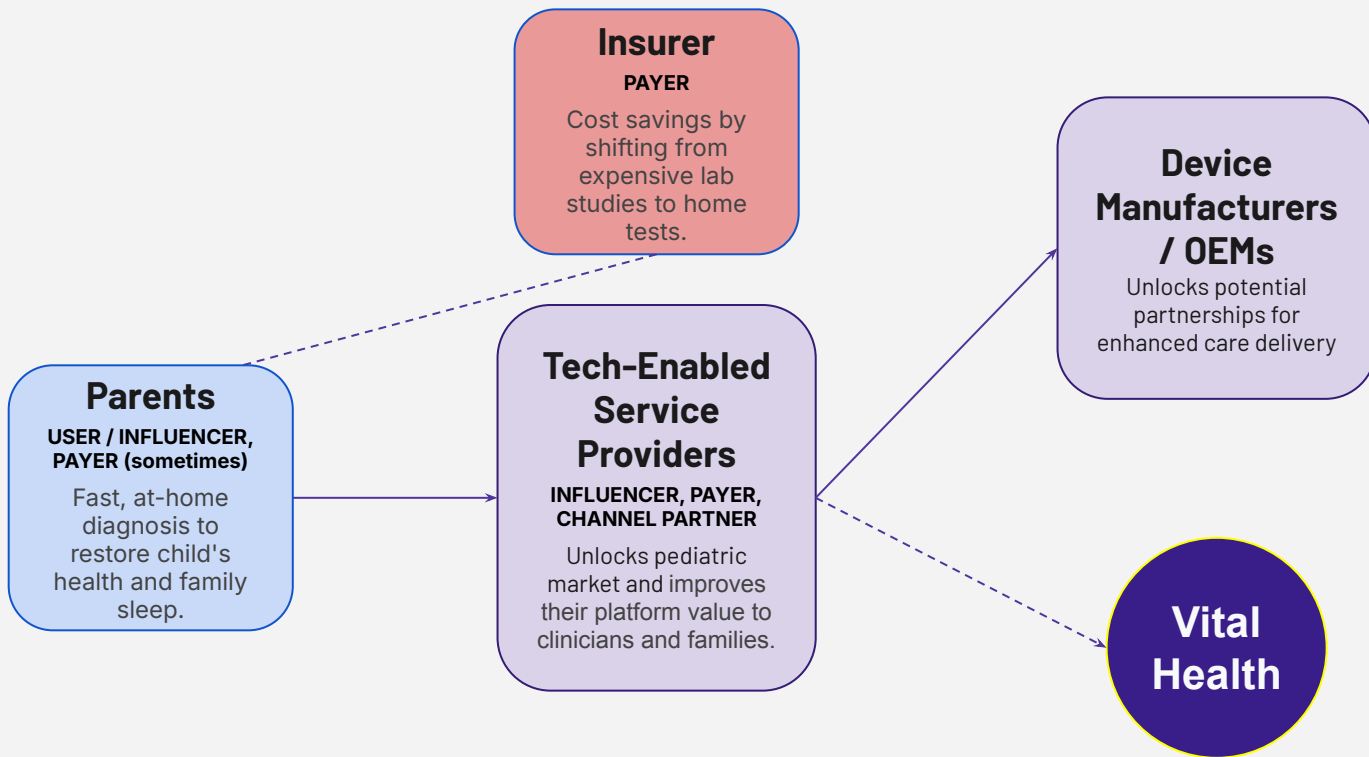
Insurer

Cost savings by shifting from expensive lab studies to home tests; long-term cost savings through earlier diagnosis

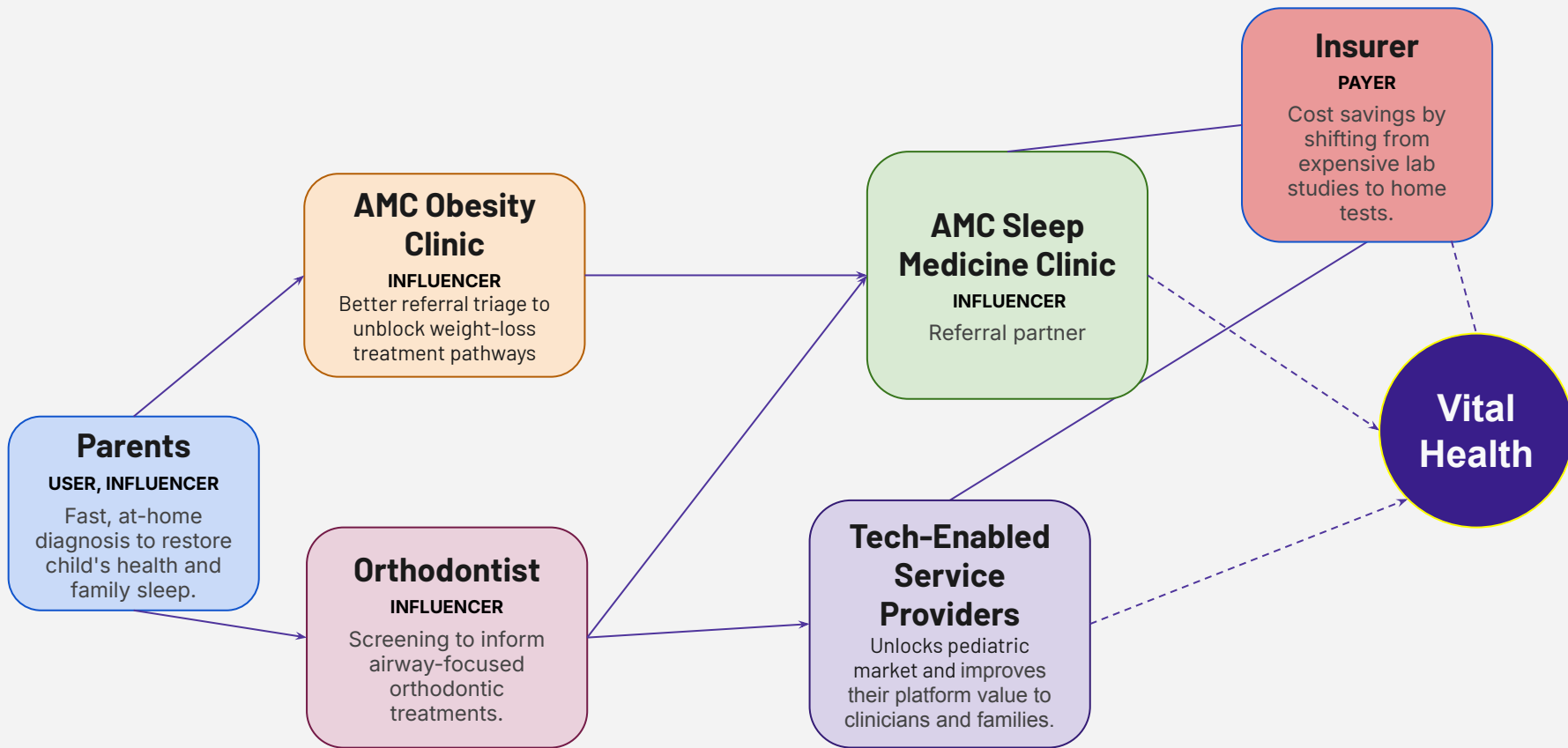
Pathway 1: AMC Sleep Medicine Clinic (B2B)



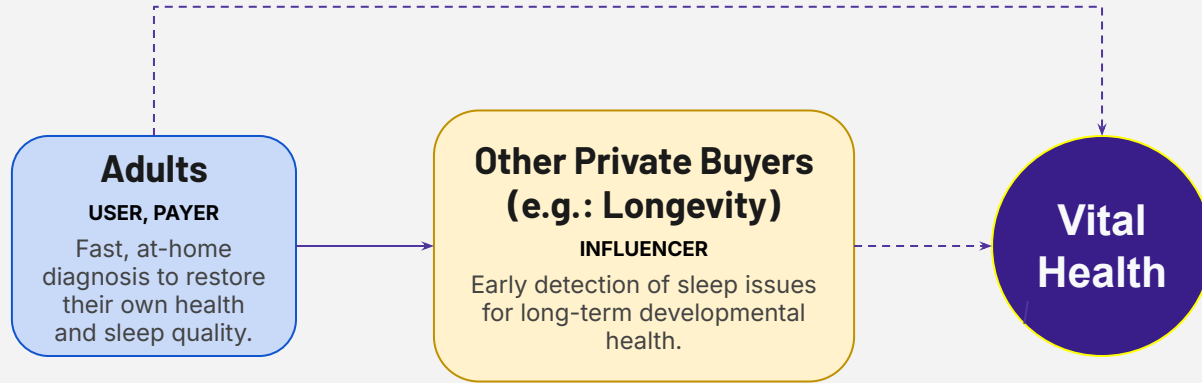
Pathway 2: Online Test Provider (B2B2C)



Pathway 3: Other Medical Specialty (B2B)



Pathway 4: Direct to Consumer (D2C)



Value Proposition (Parents)

Pain Relievers

- Reduces diagnostic wait time from 6+ months
- Eliminates need for expensive in-lab sleep studies
- Simplifies coordination by removing travel/scheduling hurdles

Gain Creators

- Provides fast clinically viable diagnostic results at home
- Reduces family stress through early health restoration

- At home pediatric OSA diagnostic device
 - Application interface for parents

Products & Services

Customer Gains

- Faster diagnosis
- Deterministic and easier process
- Relieve family stress by restoring health

Customer Pains

- Kid not sleeping well (causing behavioral problems)
- Slow diagnostic pipeline
- Lots of moving pieces (scheduling, accompaniment, coordination)

- Coordinating care for kids who are suspected of having OSA

Customer Jobs

Value Proposition (AMC Sleep Medicine Clinic)

Pain Relievers

- Provide a faster first-step test for overflow demand
- Enable home testing for children who struggle with lab studies
- Accurate and actionable data to clinicians
- Removes burden from clinicians dealing with unsatisfied families

Gain Creators

- Helps expand capacity
- Creates a clearer screening/triage wedge before full diagnostic replacement
- Improves accuracy of at-home sleep data
- Improves throughput, access, and speed to treatment

- At home pediatric OSA diagnostic device
- Application interface for clinicians (with scored and summarized sleep reports)
- Raw data streams

Products & Services

Customer Gains

- Faster screening and diagnosis
- Reliable pediatric-friendly home testing
- Better throughput without overloading lab resources

Customer Pains

- Long wait times
- Home tests often miss mild/subtle cases
- New tools face workflow resistance, reimbursement uncertainty

- Triage referred patients efficiently
- Decide home test vs. in-lab study
- Diagnose pediatric OSA accurately
- Move patients to treatment with defensible clinical data
- Fit testing into existing clinical workflow and insurer requirements

Customer Jobs

Hypotheses [1/2]



PARENTS

H1 Doctor endorsement is the primary trust signal



Eli Villa: skeptical of wellness wearables; OOP payments rely on MD rec.

H2 Parents pay \$200–\$400 OOP if clinically validated



Tyler Ogren confirmed ~\$200–300 range; Empower uses OOP superbills.

H3 Multi-child households value at-home to avoid overnight



H4 Form factor comfort is a hard requirement



Donnellan: 25–35 lab wires hurt kids; patches preferred over rings/watches.

H5 Mail-in patch model is acceptable if workflow is clean



Empower Sleep uses mail-in model with return label — works in practice.

H6 Rural families have higher hurdle tolerance



INDEPENDENT ENTs / PCPs / ORTHO

H1 In-house diagnostics meaningfully improve surgical conversion



Empower Sleep: dental/ortho act as massive B2B channels; use as pre-intervention diligence.

H2 Workflow simplicity beats clinical perfection



Curt Langlotz: piggyback existing data systems; Empower requires open APIs.

H3 CPT code is a hard prerequisite for routine use



David Kopp + Empower: can scale B2B via cash-pay/superbills before formal CPT.

H4 Physicians won't 'sell' OOP healthcare to patients



NEW: David Kopp warned relying on doctors to market OOP is a massive hurdle.

H5 Dental/Ortho is a faster-converting channel than PCPs



NEW: Tyler Onderdonk — dentists are largest untapped B2B referral channel.



OBESITY / GLP-1 CLINICS

H1 OSA screening is a hard GLP-1 insurance prerequisite



H2 Documentation compliance > clinical precision



Validated



Invalidated



New Hyp.



Not Discussed in W2

Hypotheses [2/2]



PAYORS

H1 Payors need published health-economic case before opening pathways



Not yet discussed with payor contacts.

H2 Value-based care payors more motivated than FFS payors



Not yet discussed with payor contacts.



HOSPITALS & AMCs

H1 AMCs accept startups only as 'triage' tools, not competitors



NEW: Curt Langlotz — position as initial screening to feed, not replace, hospital PSG.



TECH-ENABLED PROVIDERS & INCUMBENTS

H1 Partner with adult full-stack providers as pediatric hardware arm



NEW: Tyler Onderdonk — scale by white-labeling to existing adult IDTF/providers vs. building from scratch.

H2 License hardware/algorithms to dominant adult players



NEW: David Kopp — faster, more capital-efficient than building full-stack D2C pediatric clinic.

H3 Incumbents will license our tech for capabilities they lack



NEW: David Kopp — positional data + central vs. obstructive differentiation are major clinician differentiators.



Other channels

H1 Military/VA will adopt screening at enlistment to prevent 'pre-existing' sleep apnea disability payouts



NEW: Tyler Ogren — baseline screening at entry prevents massive downstream VA disability costs.



Validated



Invalidated



+ New Hyp.



— Not Discussed in W2



Experiments

Testing MVP and hypotheses against market reality: Testing whether our current prototype delivers the exact sensor data and workflow simplicity required to trigger adoption.

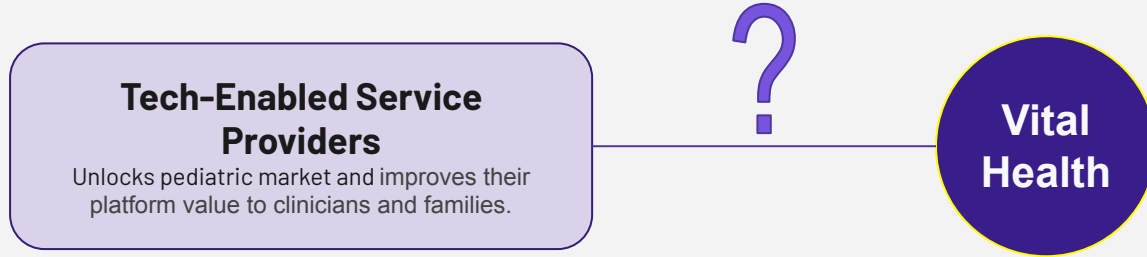
How we are testing:

- Conducting competitive analysis against existing pediatric and adult sleep testing solutions
- Comparing our MVP on: 1) Pediatric-first design 2) Multi-modal clinical sensing 3) Integrated sensing in a single device 4) AI-enabled diagnostic workflow 5) Validated in pediatric populations
- Pitching our feature set directly to independent clinics, payors, and adult market proxies (e.g., Empower Sleep) to separate the "must-haves" from the "nice-to-haves"

What the experiments validate:

- Are we a pre-PSG screening tool, a full diagnostic replacement, or a GLP-1 capacity unblocker?
- Where do we definitively outperform the competition, and what hardware gaps must we close to win?
- Does this MVP make us a full-stack service provider, or an OEM licensing to incumbents?

Experiment / MVP Demo - Value to Tech-Enabled Service Providers



Experiments run via interview:

- 1) Test attractiveness for potential partnership
- 2) Benchmark against competitors

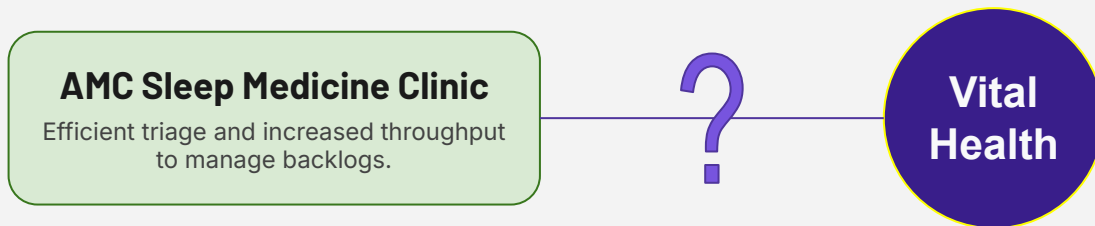
Empower Health Features

- Tech-enabled sleep testing + care platform
- Value chain: **data collection** (Sleep Image ring / Onera / others) -> **data interpretation** (Sleep image / Enso Data) -> **care management** (in-house).
- Business driven primarily by B2B provider channels
- For a device partner, important criteria are better form factor for pediatrics (ring-based devices fall off), low irritation, reliability, and easy API/SDK integration.

Potential value prop for a pediatric device:

- Better pediatric form factor (children are more active sleepers) - **they liked our form factor**
- Expand platform into high-need pediatric segment (beyond their current share)
- Offers their providers (orthodontists, GPs) a "better solution"
- Strengthens care offerings if it is easy to integrate into their software and care pipeline
- Interestingly, quality / depth of technical data was **NOT cited as a need**

Experiment - Value to Sleep Clinics



Experiments run via interview:

- 1) Validate pediatric testing pain points and unmet needs
 - 2) Test whether an at-home pediatric device is most valuable as a triage / throughput solution
- Current pediatric PSG is burdensome and wire-heavy
 - Parent discomfort is a major barrier
 - Kids often dislodge sensors
 - Labs are capacity constrained
 - A perfect study rarely occurs and the challenge is getting enough reliable data
 - Wireless monitoring is a major unmet need

Why a pediatric OSA device would be valuable:

- Reduce burden from wire-heavy, pediatric sleep tests
- Improves comfort for both children and parents
- Makes testing more feasible for active sleepers and some special-needs patients
- Helps labs manage overflow demand and long waitlists
- Could enable a screening / triage pathway before full in-lab PSG
- Creates value if it provides the key clinically important signals while simplifying setup



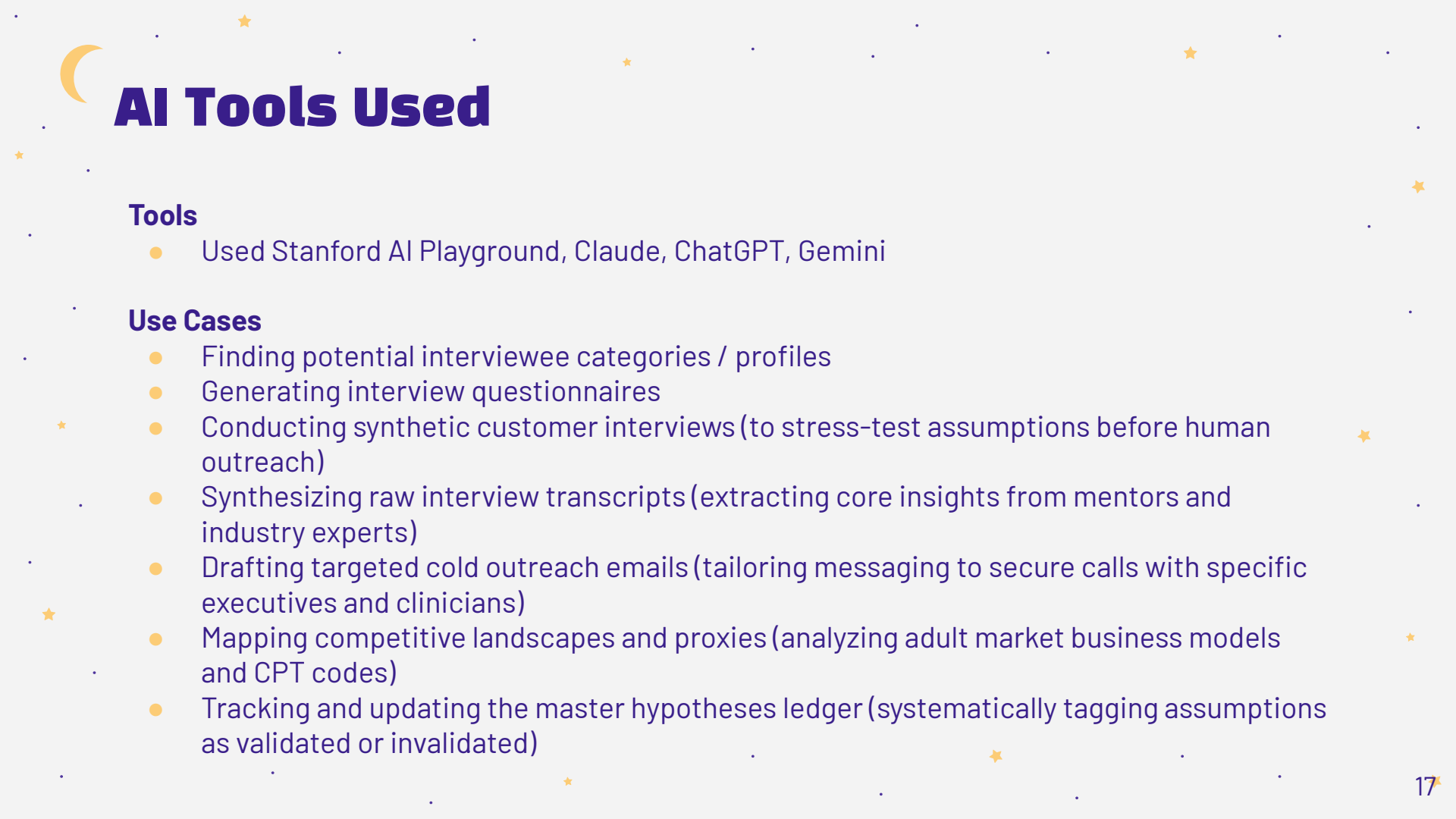
Pivot/Proceed Checkpoints

Proceed if:

- Interviews identify that our MVP solves a real unmet need and stakeholders identify a credible adoption pathway
- Our current sensor set is sufficient for the intended initial use case
- We validate a clear path to early monetization, either via out-of-pocket superbills or a direct B2B licensing partnership
- The product can fit into an existing clinical or commercial workflow without major friction

Pivot if:

- Key stakeholders indicate that our solution is not important or valuable enough to warrant adoption
- Important sensors or features are missing for adoption
- Our differentiation versus competitors is not clear
- Buyers do not see enough value
- Creates added workflow burden that is too high for clinics, intermediaries, or families
- Becomes clear that the best initial use case is different from our current target



AI Tools Used

Tools

- Used Stanford AI Playground, Claude, ChatGPT, Gemini

Use Cases

- Finding potential interviewee categories / profiles
- Generating interview questionnaires
- Conducting synthetic customer interviews (to stress-test assumptions before human outreach)
- Synthesizing raw interview transcripts (extracting core insights from mentors and industry experts)
- Drafting targeted cold outreach emails (tailoring messaging to secure calls with specific executives and clinicians)
- Mapping competitive landscapes and proxies (analyzing adult market business models and CPT codes)
- Tracking and updating the master hypotheses ledger (systematically tagging assumptions as validated or invalidated)



To-do Next Week and Help needed!

- Competitive benchmark our MVP: Compare vs. other pediatric/adult sleep tests
 - Confirm minimum clinically important signals + acceptable setup/workflow to reliably get data
- Stakeholder interviews: interview more sleep-medicine experts, companies, and other clinical stakeholders in the ecosystem to learn where our device has the best initial impact
- Learn more about the reimbursement pathway: confirm which existing codes we can qualify for now (adult at-home vs. pediatric in clinic) and outline implications for go-to-market

Help needed

- Intros to clinicians to test our hypotheses around providers, esp. across different geographies and socio-demographics
- Intros to value-based care organizations, regulatory experts, and payors
- Problem-solving (if any) around digital MVP